01 · AUTH AND ACCESS

1 · ENTER PASSWORD
user opens /login and submits the shared access password

→ **POST /api/login**
`zod-validated · JSON body`

→ **PASSWORD CHECK**
password compared against APP_PASSWORD → on match an httpOnly cookie nextcall_auth is issued (strict same-site · 7-day expiry)

→ **OUTCOME**
success → redirect to /dashboard
wrong password → 401 Invalid password

MIDDLEWARE GATE — RUNS ON EVERY /dashboard/* AND /api/* REQUEST
validates the nextcall_auth cookie → missing or wrong: pages redirect to /login · APIs return 401
/api/webhooks/* and /api/tracking/* bypass auth (svix signature + secret record id instead)

2 · LOG OUT
click logout in the dashboard sidebar

→ **POST /api/logout**

→ **COOKIE CLEARED**
auth cookie removed — session ends

→ **RETURN**
back to /login

02 · DASHBOARD HUB

LAND ON DASHBOARD
entry point after login

→ **GET /dashboard**
`server-rendered page`

→ **LIVE STATS**
total prospects · high-opportunity count (HOT / HIGH or score ≥ 55) · audits generated · outreach emails sent — read from Astra DB on every page load

→ **WHERE YOU CAN GO**
Discovery · Screenshot · Pipeline · Settings
Settings = read-only status of OpenAI · Resend · SerpAPI · Apify · TomTom · Geoapify · Astra DB · access password

03 · BUSINESS DISCOVERY

1 · SEARCH BUSINESSES
type a query + draw a map bounding box on the Leaflet map

→ **POST /api/discover**
`rate limit 5 / min`

→ **DISCOVERY ORCHESTRATOR**
providers tried in order with quota tracking + auto fallback → DeduplicationService merges duplicates into single clean records

→ **PROVIDERS + STORAGE**
SerpAPI → Apify → TomTom → Geoapify
Astra DB — deduped businesses persisted

2 · ENRICH A RESULT
click enrich on a business card to pull its contact data

→ **POST /api/enrich**
`5 / min · 30-day cache`

→ **CRAWL + EXTRACT**
HTMLFetcher crawls the website → extractors pull emails · phones · socials · booking links · site features → PriorityService recalculates the score

→ **STORED**
contact data + signals → Astra DB
priority score recalculated after enrichment

3 · RUN THE AUDIT
one click on the enriched business

→ **POST /api/audit**
`5 / min · 30-day cache`

→ **AUDIT ENGINE + AI**
deterministic AuditEngine scores digital presence (30-day cache) → AI report drafted via OpenAI → both stored · AI failure still returns the deterministic audit

→ **STORED**
deterministic audit + AI report → Astra DB
priority recalculated from fresh findings

4 · REPORT + ADD TO PIPELINE
view the audit PDF, then push the business into your pipeline

→ **POST /api/reports/generate**
`POST /api/prospects`

→ **REPORT + PROSPECT**
reportService renders the audit as a PDF (@react-pdf/renderer) · OpportunityService scores the business → prospect created with sequence state UNCONTACTED

→ **OUTPUT**
audit PDF served inline in the browser
prospect lands in the Pipeline

04 · SCREENSHOT INTAKE

1 · UPLOAD A SCREENSHOT
Google Maps listing image, ≤ 8 MB

→ **POST /api/extract-screenshot**
`6 / min · image only`

→ **AI VISION EXTRACTION**
gpt-4o-mini reads the listing → name · phone · website · address · rating · reviews · category · email — only what is clearly visible; the rest is listed back as missing

→ **AI + LIMITS**
OpenAI gpt-4o-mini (vision)
missing fields reported back for manual completion

2 · REVIEW + CONFIRM
fix fields, fill gaps, add notes (socials · booking · website)

→ **POST /api/businesses/manual**
`5 / min`

→ **MERGE + AUDIT**
notes parsed for user-verified signals → business upserted → website crawled → verified signals merged over crawl findings (never overwritten) → audit + AI report → priority

→ **STORED + NEXT**
business + enrichment + audit → Astra DB
then reuse VIEW REPORT + ADD TO PIPELINE, same endpoints as Discovery

05 · PIPELINE AND CRM

1 · OPEN THE PIPELINE
filter by sequence status

→ **GET /api/prospects?status=**

→ **RANKED LIST**
prospects sorted by priority score (highest first) each row joined with business name + latest audit score

→ **LIST READY**
HOT / HIGH opportunities surface first

2 · OPEN A PROSPECT
detail view

→ **GET /api/prospects/[id]**
`view is recorded`

→ **FULL CONTEXT**
prospect + business + latest audit + AI report + full outreach history · missing business doc is rebuilt from the stored snapshot

→ **DETAIL READY**
everything needed to decide the next move

3 · MANAGE THE PROSPECT
status (8 stages) · priority level · contact email · suppress + reason · notes (parsed for keyword signals) · next follow-up date · delete

→ **PATCH /api/prospects/[id]**
`DELETE /api/prospects/[id]`

→ **APPLY + CASCADE**
PATCH → granular field updates applied · DELETE cascades: outreach history + all audits + the business document itself

→ **EFFECT**
suppressed prospects can never be emailed
delete removes every trace from Astra DB

06 · OUTREACH ENGINE

1 · SEND INITIAL EMAIL
one click on the prospect

→ **POST /api/outreach**
`10 / min`

→ **SAFETY → BUILD → CLAIM → SEND**
suppression · sequence state · email validity checked → AI audit required → email built from audit + AI report with the audit PDF attached → pending record + open-tracking pixel → ATOMIC CLAIM UNCONTACTED → INITIAL_SENT (concurrency guard)

→ **SEND + RESULT**
Resend API · idempotency key per prospect + stage
success → record marked sent + priority recalc
failure → sequence state reverted, record marked failed

2 · DRAFT A FOLLOW-UP
AI writes it, you keep control

→ **POST /api/outreach/followup-draft**

→ **AI DRAFT**
stage-aware draft (FU-1 → FU-3 → close-the-loop) grounded in business data, the AI report and previous emails

→ **DRAFT READY**
draft returned to the UI for editing
nothing is sent until the user approves

3 · EDIT + SEND FOLLOW-UP
stage 1→4 · stage 4 closes the loop

→ **POST /api/outreach/send-followup**
`10 / min`

→ **ATOMIC STAGE SEND**
same safety + idempotency as the initial send · atomic state transition per stage · stage 4 = CLOSE_LOOP_SENT · next follow-up scheduled +7 days on the manual path

→ **SEQUENCE ADVANCED**
state machine: INITIAL_SENT → FOLLOW_UP_1/2/3_SENT → CLOSE_LOOP_SENT

07 · AUTOMATION + TRACKING

RECIPIENT OPENS EMAIL
outside the app — no login

→ **GET /api/tracking/open?r=**
`public · no auth`

→ **PIXEL PING**
1×1 transparent GIF served instantly → open timestamp written to the outreach record (fire-and-forget — a tracking failure never blocks the pixel response)

→ **OPEN RECORDED**
opens drive the automated follow-up timing

EMAIL EVENTS ARRIVE
delivered · bounced · complained

→ **POST /api/webhooks/resend**
`public · svix-signed`

→ **SIGNATURE-VERIFIED EVENT**
svix signature + 5-minute replay window → delivery status written to the outreach record → bounce or complaint auto-suppresses the prospect (no future sends)

→ **STATUS + SAFETY**
delivered / bounced / complained stored · auto-suppression

NO USER ACTION
background automation

→ **SCHEDULER SWEEP**
`every 15 min from boot`

→ **AUTO FOLLOW-UP SWEEP**
FU-1 fires 3 days after the initial email is opened · FU-2 fires 7 days after open if FU-1 was opened → AI draft (template fallback) → same atomic send pipeline

→ **SEQUENCE CONTINUES**
only opened + delivered emails earn a follow-up

USER ACTION · APP PAGE · API ENDPOINT
AUTOMATED · EXTERNAL SYSTEM · PUBLIC ROUTE
REQUEST / DATA FLOW
n / min = API RATE LIMIT
NUMBERS = STEP ORDER WITHIN A FEATURE

NEXTCALL MARKETING APP · COMPLETE USER ACTIVITY AND SYSTEM FLOW · DISCOVERY → SCREENSHOT → PIPELINE → OUTREACH → AUTOMATION